

Task Manager V6 Master Blueprint

Purpose

V6 introduces Search, Filter, and Sort. The goal is to make larger task lists easier to navigate and organize without changing how tasks are stored. Unlike V5, this version focuses on functionality and task views rather than adding new task fields.

Version Goal

Add:

- Search
- Filter
- Sort

while preserving all V1-V5 functionality.

Why V6 Exists

After V5, tasks contain:

- Name
- Completed Status
- Priority
- Category

As task counts grow, users need ways to find, group, and organize tasks quickly.

Architecture

Maintain:

Main

↓

TaskManager

↓

Task

Main:

- Menus
- Input
- Output

TaskManager:

- Search
- Filter
- Sort

Task:

- Stores task data

Search Design

Search only checks task names.

Does NOT search:

- Priority
- Category
- Completion Status

Reason:

Filtering already handles those fields.

Search Rules

- Partial matching
- Case-insensitive
- Display-only
- Returns ArrayList

Examples:

Search: java

Matches:

Study Java
Java Homework
Learn Java OOP

Search: ava

Matches:

Study Java
Java Homework

Search Validation

If task list is empty:

No tasks available.

If no results:

No matching tasks found.

Blank search terms are not allowed.

Filter Design

Filter Menu:

1. Filter by Priority
2. Filter by Category
3. Filter by Completion Status

Reason:

Cleaner menu and easier expansion.

Priority Filter

Options:

High

Medium

Low

Displays matching tasks only.

Category Filter

Options:

School

Work

Personal

Fitness

Church

Other

Displays matching tasks only.

Completion Filter

Options:

Completed

Incomplete

Displays matching tasks only.

Filter Behavior

Filtering:

- Creates result list
- Returns ArrayList
- Does not modify tasks
- Does not reorder tasks
- Does not delete tasks

Sort Design

Sort Menu:

1. Sort by Priority
2. Sort by Completion Status

Display-only feature.

Original task order never changes.

Priority Sorting Decisions

Priority Values:

High = 3

Medium = 2

Low = 1

Reason:

Higher number = higher priority.

Sort Order:

High

↓

Medium

↓

Low

Completion Sorting Decisions

Sort Order:

Incomplete

↓

Completed

Reason:

Users usually want to see unfinished work first.

Sort Implementation Decisions

1. Use same Task objects
2. Create copy of task list
3. Sort copy
4. Return copy

Never sort the original task list.

Tie Handling

If two tasks have the same priority:

Keep original order.

If two tasks have the same completion status:

Keep original order.

Helper Methods

TaskManager owns sorting logic.

Example helper:

getPriorityValue(String priority)

High -> 3

Medium -> 2

Low -> 1

Unknown -> 0

Reason:
Sorting logic belongs in TaskManager.

Sort Validation

If task list is empty:

No tasks available.

Check before displaying sort menu.

TaskManager Methods

Search:
searchTasks(String searchTerm)

Filters:
filterByPriority(String priority)
filterByCategory(String category)
filterByCompletionStatus(boolean completed)

Sorts:
sortByPriority()
sortByCompletionStatus()

Return Types

All Search, Filter, and Sort methods return:

ArrayList

Reason:
TaskManager provides data.
Main handles presentation.

Reusable Display Method

Create:

displayTasks(ArrayList tasks)

Used for:

- View Tasks
- Search Results
- Filter Results
- Sort Results

Reduces duplicate loops.

Result Formatting

Task Format:

Study Java [] (High) [School]

Headers:

Search Results:

High Priority Tasks:

School Tasks:

Tasks Sorted By Priority:

Tasks Sorted By Completion Status:

Results should always be numbered.

Menu Layout

1. Add Task
2. View Tasks
3. Search Tasks
4. Filter Tasks
5. Sort Tasks
6. Complete Task
7. Delete Task
8. Rename Task
9. Mark Task Incomplete
10. Set Task Priority
11. Set Task Category
12. Exit

Important Limitation

Search, Filter, and Sort are viewing features only.

Users cannot directly edit tasks from search/filter/sort results because result numbers may not match actual task numbers.

Potential future version feature.

File Format

No file format changes.

Continue using:

TaskName,completed,priority,category

Example:

Study Java,false,High,School

Testing Checklist

Search:

✓ Partial matching

- ✓ Case-insensitive
- ✓ Empty results
- ✓ Empty task list

Filter:

- ✓ Priority
- ✓ Category
- ✓ Completion Status

Sort:

- ✓ Priority order
- ✓ Completion order
- ✓ Original order unchanged

Regression:

- ✓ Add Task
- ✓ View Tasks
- ✓ Complete Task
- ✓ Delete Task
- ✓ Rename Task
- ✓ Mark Incomplete
- ✓ Set Priority
- ✓ Set Category
- ✓ Save Tasks
- ✓ Load Tasks

V6 Complete When

- ✓ Search works
- ✓ Filter works
- ✓ Sort works
- ✓ Results are numbered
- ✓ No matching tasks found works
- ✓ Empty task list handled
- ✓ Original task order unchanged
- ✓ Save/load works
- ✓ All V1-V5 features work
- ✓ Changes committed to GitHub

Future Version Notes

Possible V7:

- Refactoring
- Cleaner Main class
- Cleaner TaskManager
- Reusable validation helpers
- Interactive search/filter results
- Better architecture cleanup